

Functional & Performance Testing Template

Model Performance Test

Date	13 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer (VBCUA)
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass / Fail
FT-01	Audio File Format Validation	Upload valid audio files (WAV, MP3, M4A) and an invalid file type (e.g., .txt, .avi) through the uploader.	Valid formats are accepted and processed; invalid formats are rejected by the file uploader before submission.	The uploader restricted selection to WAV/MP3/M4A as configured; files outside these types could not be selected for upload.	Pass
FT-02	Empty / Corrupted Audio File Handling	Upload an empty or corrupted audio file and attempt transcription.	The system displays a descriptive transcription error instead of crashing.	A “Transcription failed” warning was shown via the TranscriptionError handling in transcriber.py; the app remained responsive.	Pass
FT-03	Speech-to-Text Transcription	Upload a clear spoken explanation of a concept and compare the generated transcript with the spoken content.	The transcript reasonably reflects what was spoken, with minor errors acceptable for domain-specific terminology.	Transcript closely matched the spoken explanation; a small number of technical terms were mis-transcribed, consistent with the Whisper “base” model's known accuracy trade-off.	Pass
FT-04	Missing / Empty Reference Answer Handling	Rename or empty the reference answer file, then run an analysis.	The system shows a warning that the reference answer could not be loaded and skips scoring gracefully.	A warning (“Reference answer file could not be found” / “could not be loaded”) was displayed as implemented in main.py; no crash occurred.	Pass
FT-05	Semantic Similarity & Score Calculation	Submit transcripts with high, medium, and low conceptual overlap with the reference answer and observe the similarity score.	The similarity score increases with conceptual overlap between the transcript and the reference answer.	Closely matching explanations scored above 85%, partially correct explanations scored 60–75%, and off-topic explanations scored below 40%, consistent with cosine similarity behaviour.	Pass
FT-06	Grade & Feedback Generation	Trigger scores at and around each grade boundary (90%, 75%, 60%) and verify the assigned grade and feedback text.	Grade tiers (Excellent / Good / Fair / Needs Improvement) and feedback text match the thresholds defined in concept_scorer.py.	Grade and feedback text matched the expected tier at each boundary value tested.	Pass

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FT-07	PDF Report Generation	Complete a full analysis (upload, transcribe, score) and confirm a PDF report is generated.	A PDF report is created containing the transcript, similarity score, grade, feedback, and generation timestamp.	A PDF report with all required sections was generated and saved to reports/concept_report.pdf.	Pass
FT-08	Report Download	Click “Download PDF Report” after a successful analysis.	The generated PDF downloads correctly with the expected file name (concept_report.pdf).	The download button correctly served the generated PDF file.	Pass
FT-09	Missing Report Handling	Load the application and attempt to download a report before any audio has been analysed.	The download action is disabled with a prompt to analyse an audio file first.	A disabled button reading “Generate a report by analyzing an audio file.” was shown, as implemented in report_actions.py.	Pass
PT-01	Transcription Response Time	Use a timer to measure how long Whisper takes to transcribe a ~1-minute audio clip on typical local hardware (CPU).	Transcription should complete within an acceptable time relative to audio length (indicative target: under ~20 seconds for a 1-minute clip on CPU).	Transcription completed in approximately 12–18 seconds for a 1-minute clip on a standard CPU-only machine; timing varies with audio length and hardware.	Pass
PT-02	End-to-End Analysis Time	Measure the total time from audio upload to results being displayed (transcription + similarity computation + scoring).	The full pipeline should complete within a reasonable time for classroom use (indicative target: under ~30 seconds for a 1-minute clip).	End-to-end processing completed in approximately 15–22 seconds, dominated by the transcription step; semantic similarity and scoring added negligible additional time.	Pass
PT-03	Model Loading & Caching Test	Run multiple analyses within the same session and confirm the Whisper and Sentence Transformer models load only once.	Models are loaded once and cached in memory; subsequent analyses in the same session skip the model-loading step.	The first analysis included noticeable model-loading time; subsequent analyses in the same session were faster, confirming the caching behaviour in transcriber.py and semantic_analyzer.py.	Pass
PT-04	Repeated Upload Load Test	Upload and analyse several different audio files sequentially within a single session.	The application processes each file without crashing, memory errors, or degraded responsiveness.	All sequential uploads were processed successfully; no crashes or memory issues were observed during repeated use within a session.	Pass